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HAOWEN ZHANG

Reinforcement Learning & Intelligent Decision-Making | Autonomous Navigation / Planning / Multi-Agent Coordination / Edge Deployment

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PROFILE

Ph.D. candidate in Information and Communication Engineering at Wuhan University, focusing on depth-vision temporal decision-making, reinforcement learning, multi-agent coordination, and generative trajectory policies. Published a first-author paper on end-to-end UAV autonomous navigation at IEEE ICRA, a top-tier robotics conference. End-to-end experience in POMDP/Dec-POMDP formulation, PPO/MAPPO design, imitation learning and DAgger data loops, ablation studies, Sim-to-Real transfer, and real-world UAV validation. Independently develops Isaac Sim environments and dynamics, training pipelines, and ROS 2/TensorRT edge systems. Previously spent two years at Huawei on BiSheng/LLVM and AArch64 performance engineering, with strong C/C++ compiler optimization, cross-layer profiling, edge performance tuning, and complex system debugging skills.

CORE SKILLS

Reinforcement Learning & Planning | End-to-end decision-making, trajectory prediction and planning, multi-agent coordination, imitation learning, generative policies; PPO/MAPPO

Autonomous Robotics | UAV navigation, depth-only obstacle avoidance, motion control, multi-sensor temporal fusion, Sim-to-Real, SIL/HIL, and real-world validation

Training & Simulation | PyTorch, Isaac Sim/Lab, environment and dynamics modeling, curriculum learning, domain randomization, data loops, evaluation, and ablation studies

Engineering & Edge Deployment | Python, C/C++, Linux, ROS 2, PX4, TensorRT, Jetson Orin NX; system integration, real-time inference, profiling, and complex troubleshooting

RESEARCH PROJECTS

End-to-End UAV Autonomous Navigation with Depth Vision and Reinforcement Learning

TECH STACK | Depth-Only Perception / DCE / Temporal Self-Attention / PPO / Isaac Sim / ROS 2 / TensorRT

- **CONTEXT |** Developed a map-free point-to-point navigation and obstacle-avoidance policy for low-altitude UAVs in warehouse inspection and search-and-rescue settings. The agent receives only the target's relative position and depth images as its sole exteroceptive input.
- **CHALLENGES |** 1) A single depth frame lacks temporal cues for occlusion changes and motion trends; 2) high-speed flight couples control delay and inertia, so decisions must use visual history, vehicle state, and prior actions; 3) asynchronous depth/VINS streams, sensor noise, dynamics mismatch, and limited onboard compute must be handled while sustaining an approximately 50 Hz Sim-to-Real control loop.
- **DEPTH REPRESENTATION |** Designed a collision-aware depth encoder (DCE) that compresses depth images into compact features preserving obstacle distance, spatial structure, and collision risk. Continued training its reconstruction/collision objectives on real UAV depth data, then froze the adapted encoder for policy learning and deployment.
- **TEMPORAL DECISION-MAKING |** Fused the current depth feature, VINS pose/state, and previous action into each time-step token; applied causal temporal self-attention over a history window so each decision exploits visual changes, action history, and state feedback, improving partial-observability and inertia modeling.
- **RL & SIM-TO-REAL |** Independently built Isaac Sim scenes, UAV dynamics, and the PPO training pipeline; shaped rewards for progress/arrival, collision, control smoothness, orientation, and hover; applied an obstacle-density curriculum and domain randomization over mass, inertia, thrust response, control latency, and depth noise to reduce the Sim-to-Real gap.
- **EDGE DEPLOYMENT |** Validated the flight controller and companion computer with HIL; accelerated policy inference on Jetson Orin NX using TensorRT. Implemented ROS 2 ApproximateTime synchronization plus VINS-buffer interpolation for heterogeneous sensor rates, enabling approximately 50 Hz real-UAV closed-loop control.
- **RESULTS |** Achieved 71.13% navigation success, 7.68 percentage points above a GRU temporal baseline, and 86.0% path efficiency in environments with 500 obstacles. Independently delivered the algorithm, simulation, training, and real-world deployment, resulting in a first-author IEEE ICRA paper.

Huawei Optical Product Line President's Award

MULTI-AGENT RL & GENERATIVE DECISION-MAKING

Decentralized Multi-UAV Formation Maintenance under Communication Loss

TECH STACK | MAPPO / CTDE / Causal Temporal Attention / Chebyshev Spectral Graph / OmniDrones

- **CONTEXT |** Addressed coordination degradation caused by unreliable wireless links in swarm inspection and cooperative search. Built a seven-UAV fixed-formation acquisition and maintenance task in Isaac Sim/OmniDrones with joint constraints on target geometry, centroid, heading, and inter-agent safety; each UAV executes solely from ego-centric history.
- **CHALLENGES |** 1) Random directed-link loss makes neighbor data missing or stale and turns formation maintenance into a time-varying Dec-POMDP; 2) CTDE creates an information gap between a globally informed critic and locally informed actors; 3) multi-hop coordination on dynamic graphs must avoid per-step O(N^3) eigendecomposition while retaining scalability and interpretable convergence.
- **ALGORITHM |** Developed an asymmetric MAPPO actor-critic under CTDE: the centralized critic fuses joint observation histories, geometric membership, and multi-scale spectral graph propagation for stable value estimation and team credit assignment; a parameter-shared actor uses only local causal temporal features for leaderless execution.
- **EFFICIENT PROPAGATION & THEORY |** Approximated weighted spectral graph propagation with Chebyshev polynomials, reducing per-step complexity from O(N^3) to O(K|E|d_h). Analyzed hidden coordination signals as dynamic information flow and derived a sufficient condition for exponential contraction under per-step connected interaction graphs.
- **COMMUNICATION EVALUATION |** Trained with nominal communication, then independently injected random loss into every directed link at test time and held the latest valid message. Evaluated degradation and computational scalability across loss rates, swarm sizes, and graph structures; retained 73% of nominal return at 40% packet loss.
- **RESULTS |** Against MAPPO under nominal communication, improved average return by 56.0%, reduced centroid-position error by 50.8%, and cut convergence steps by 37.0%. Completed multi-baseline studies, nominal-condition ablations, and scalability tests from 7 to 28 UAVs; prepared a first-author IEEE TNNLS manuscript under review.

Distributed Multi-UAV Formation Navigation with Conditional Flow Matching and Reinforcement Learning

TECH STACK | Conditional Flow Matching / DAgger / Quality-Diversity MMR / Masked-Joint MAPPO / Distributed Planning

- **CONTEXT |** Built a distributed three-UAV formation-navigation system for local depth sensing and limited communication. The swarm maintains a recoverable triangular formation, elastically deforms to bypass a static obstacle that fully blocks the direct path, restores formation, then decelerates and holds at a shared goal.
- **CHALLENGES |** 1) The distributed CFM must preserve multimodal futures such as left/right bypass, deceleration, and recovery, while locally feasible plans can conflict jointly; 2) fixed two-round communication and Top-M truncation must balance candidate quality, diversity, joint safety, and bounded latency; 3) post-hoc coordination overrides would break PPO's on-policy semantics, so the executed joint tuple must itself be an action with tractable log-probability; 4) closed-loop replanning requires deterministic fallback and real-time execution.
- **GENERATIVE POLICY & DATA LOOP |** Independently built the Isaac Sim task, centralized geometric joint expert, and demonstration pipeline. Applied distributed-policy DAgger to relabel collision, deviation, and terminal-recovery states visited by the learned policy, then trained a parameter-shared distributed conditional flow model to generate 2 s multimodal high-level trajectory candidates from structured local history and anonymous neighbor messages.
- **RL-BASED JOINT DECISION |** Froze the flow model and retained a quality-diverse Top-8 from 32 candidates using near-term actions and long-horizon trajectory shape. Designed a parameter-shared masked-joint MAPPO ranker that combines centroid-intent consensus with fixed two-round communication to select the feasible joint tuple maximizing long-horizon team return; the centralized critic is used only during training.
- **TRAINING OBJECTIVE |** Formulated each replan as a semi-MDP macro-step and reused an additive team reward covering target progress, obstacle/inter-UAV clearance, formation deformation/recovery, smoothness, and goal holding; added only an emergency-braking penalty when no feasible joint tuple exists. All agents share team reward and advantage.
- **SAFETY & REAL-TIME LOOP |** In the second communication round, exchanged sparse trajectory nodes and deterministically enumerated joint tuples. Applied hard masks for inter-UAV clearance, local validity, and terminal formation; executed a fixed emergency-braking tuple when none is feasible, replanned after the first two high-level actions, and stored rollout candidates/masks to preserve PPO on-policy support.
- **DEPLOYMENT & OUTPUT |** Completed distributed closed-loop validation on three Sunray300 UAVs, integrating local perception, neighbor communication, trajectory generation/ranking, elastic obstacle avoidance, formation recovery, and goal-region hold. Independently delivered simulation, expert data, model training, RL optimization, system deployment, and manuscript preparation, producing a completed research paper.

PROFESSIONAL EXPERIENCE

Huawei Technologies | High-Performance Computing Engineer Sep 2021 - Jul 2023

BiSheng/LLVM / C/C++ / AArch64 / Build & Runtime Profiling

- **BUSINESS CONTEXT |** Supported optical-transport and optical-sensing workloads, including indoor human-activity sensing and fiber perimeter security, by deploying the BiSheng/LLVM toolchain to AArch64 edge devices. Targeted build efficiency, runtime performance, binary footprint, and long-term stability of large C/C++ software.
- **CORE ROLE |** Served as a core technical contributor for workload assessment, metric definition, GCC-to-BiSheng/LLVM migration, and system-level optimization. Established performance baselines and regression checks across compilation, linking, loading, and runtime, and resolved critical performance bottlenecks.
- **TECHNICAL EXECUTION |** Built a cross-layer profiling workflow spanning source/dependencies, compiler IR, object/link stages, and AArch64 hardware execution. Applied compiler-flag and dependency pruning, dead-code elimination, tiered LTO/ThinLTO, symbol/debug-info stripping, parallel builds, and compatibility fixes, including undefined behavior exposed by stricter LLVM diagnostics.
- **IMPACT & TRANSFER |** Reduced compile time by 15%, improved runtime efficiency by 10%, and cut binary size by 25%; received the Huawei Optical Product Line President's Award. Reused the same profiling and optimization discipline for TensorRT policy inference and ROS 2 real-time deployment on Jetson Orin NX.

SELECTED PUBLICATIONS & IP

- **FIRST AUTHOR |** STAF-Navi: Vision-Based Spatio-Temporal Attention Fusion Navigation Framework, IEEE International Conference on Robotics and Automation (ICRA)
- **FIRST AUTHOR |** Learning Distributed Coordination on Dynamic Information Manifolds: A Geometric MARL Framework for Aerial Swarms, under review at IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- **CO-AUTHOR |** HiDeGS, International Journal of Applied Earth Observation and Geoinformation, 150:105336, 2026; Dual-Stream UNet, IEEE Transactions on Geoscience and Remote Sensing, 64:1-15, 2026
- **INTELLECTUAL PROPERTY |** Contributed to or filed 7 patent applications related to end-to-end autonomous navigation and multi-UAV navigation, localization, and task planning

EDUCATION

Wuhan University | Ph.D. Candidate Sep 2023 - Jun 2027 (Expected)

Information and Communication Engineering | 2025 National Doctoral Scholarship; First Prize, Zhixing Forum Oral Presentation

Beijing University of Posts and Telecommunications | M.Eng. Sep 2019 - Jun 2022

Electronic and Information Engineering | Outstanding Master's Graduate; First-Class Scholarship for three consecutive years

Lanzhou Jiaotong University | B.Eng. Sep 2015 - Jun 2019

Communication Engineering | First-Class Scholarship for four consecutive years; Excellence Award, Huawei ICT Competition